



3DEXPERIENCE®

NAM Service Accelerator(LPM)

September, 2021

LPM Update Agendas

▶ LPM Landscape

▶ LPM new Features

- ▷ PPR screen

- ▷ Execution

- ▷ NCR and Order Cockpit

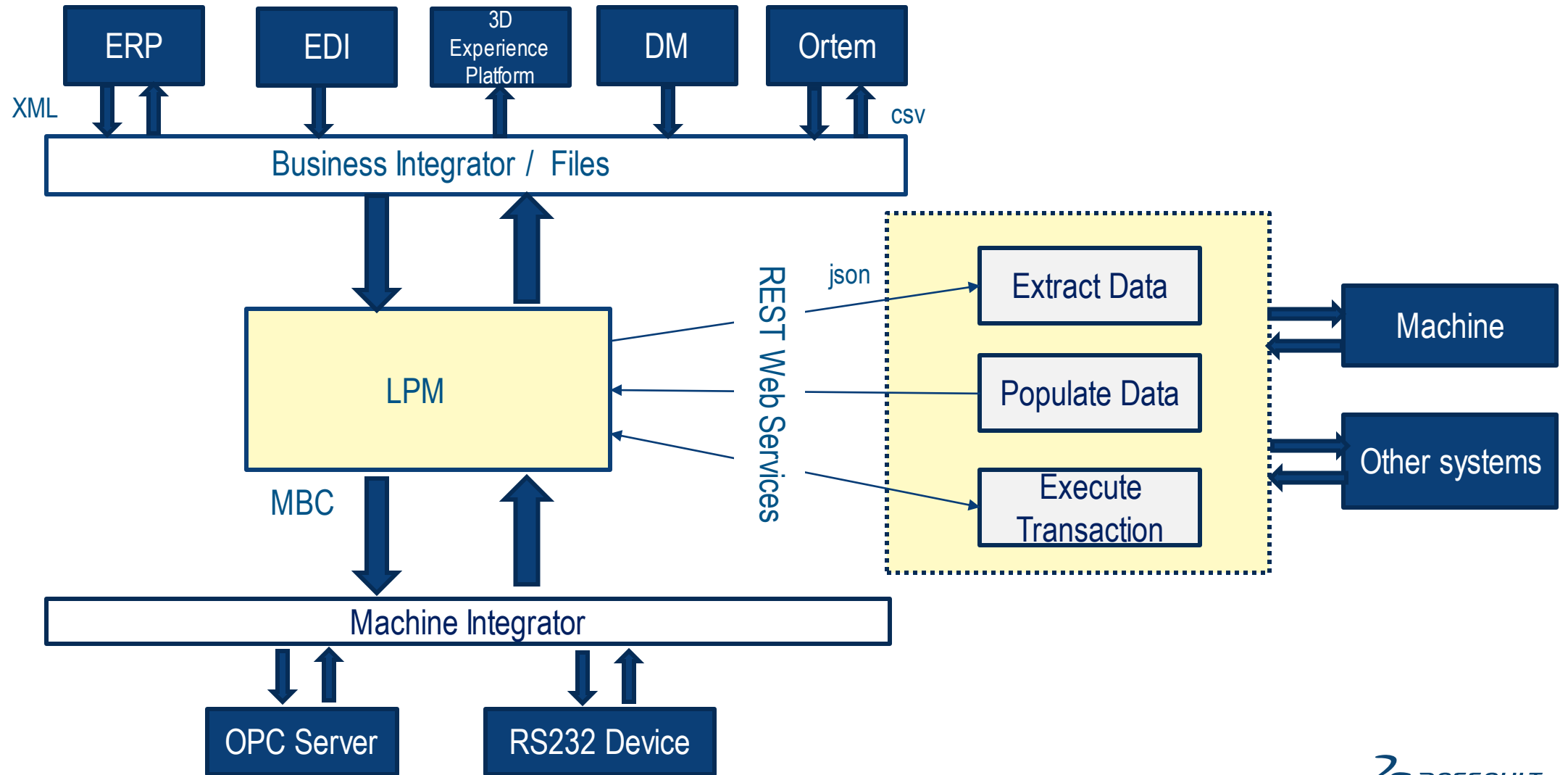
▶ LPM Evolution / Roadmap

High Level LPM functional Landscape

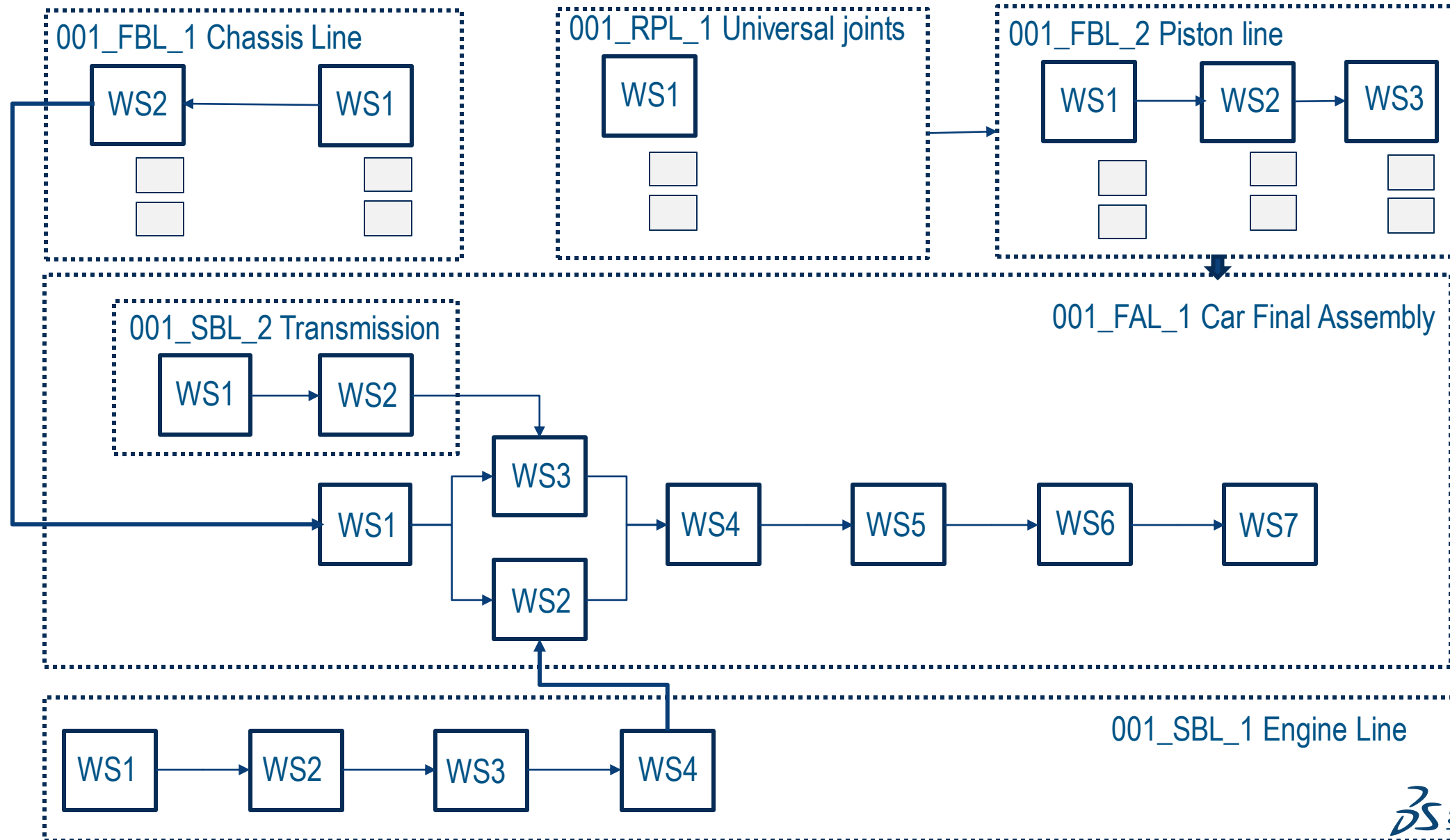
LPM OOB / New Features
LPM OOB
Platform/Apriso OOB
LPM Partial / in process
LPM future

Integration	Standard schema for ERP Production order Standard schema for EDI Broadcast Message Integration with Ortems Integration with DM OOB REST Web Services									
KPIs	KPI Cockpits (more than 20 KPIs) More than 20 Reports (Cycle time, Line OEE, TEEP etc.) Genealogy Lookup and Comparison Work Order History and EMR MPI and MPI Lite PMC MTA Factory Maps (2D & 3D)									
Process	PPR from DM Process in Apriso PPR screen PLS PPLS MBC ECN Order and Queue Mngt Planning Cockpit									
Materials	Materials Trigger Kitting Order Execution Wip Container Tracking Materials Tracking & Visibility (2D & 3D) Inventory Consumption Integration with WMS (Inventory) E-Kanban									
Use Cases	Assembly Production Fabrication Production Repetitive Production Machine Driven Production (QPM) Continuous Production Nomad (Offline mode) NCR Alerting CAPA Flow Maintenance Time & Labor									
Functions	Materials Tools Work Instruction 3D WI by Operation/Step CheckList & Buyoff Inspection Data Collection (DM) VQDT (2D/3D) Notice (DM Alert) SignOff (DM & Apriso) Rework Labeling									
	Archiving Licensing Online Help document & user guide Common Library & Standard UI Style UI theme and configurable Head Automatic Test solution OOTB Test Case Excel Data Loader Report the LPM issues									

LPM Integration Landscape



LPM Virtual Digital Factory Demo



LPM Update Agendas

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▶ LPM new Features

- ▷ PPR screen

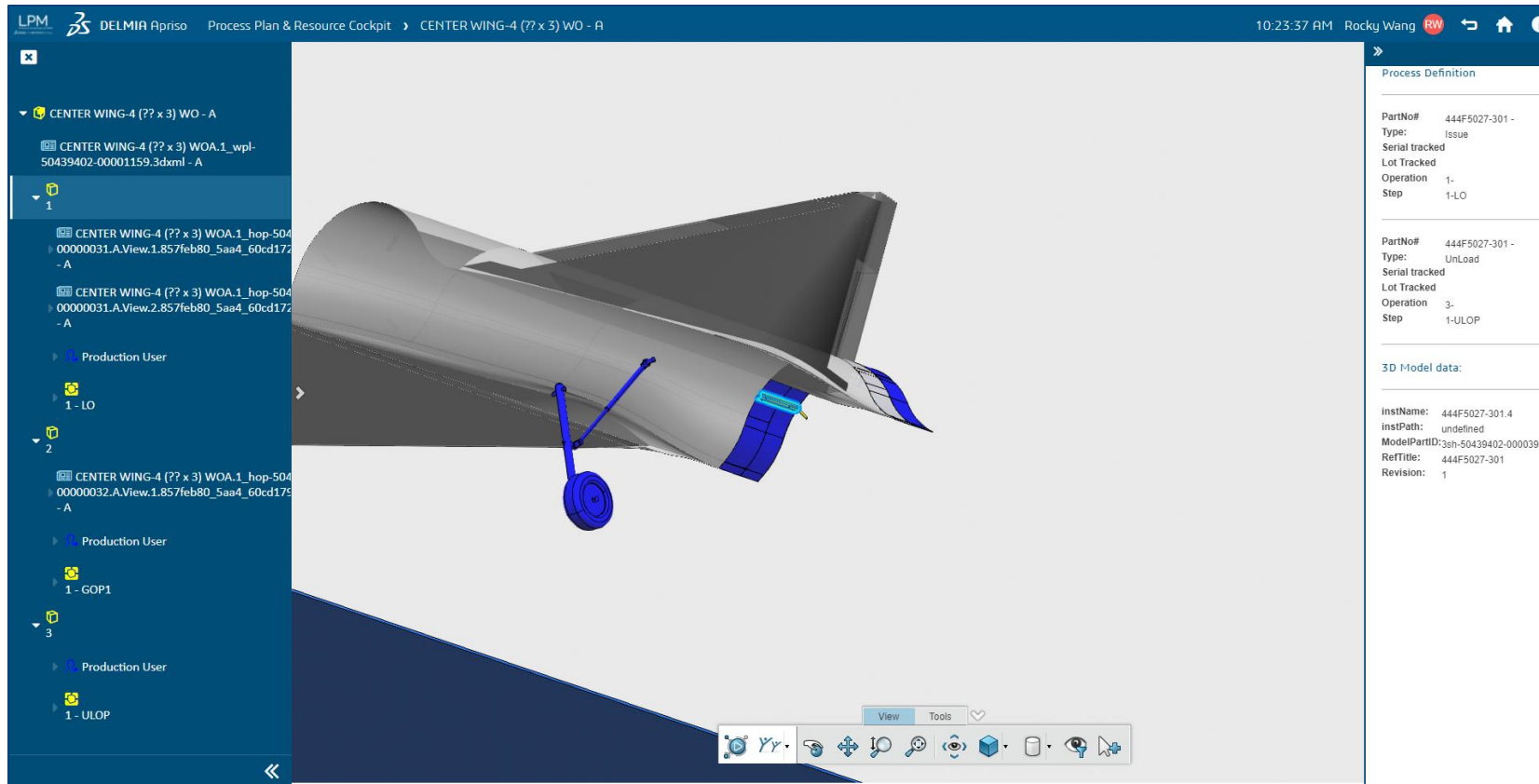
- ▷ Execution

- ▷ NCR and Order Cockpit

▶ LPM Evolution / Roadmap

New PPR screen

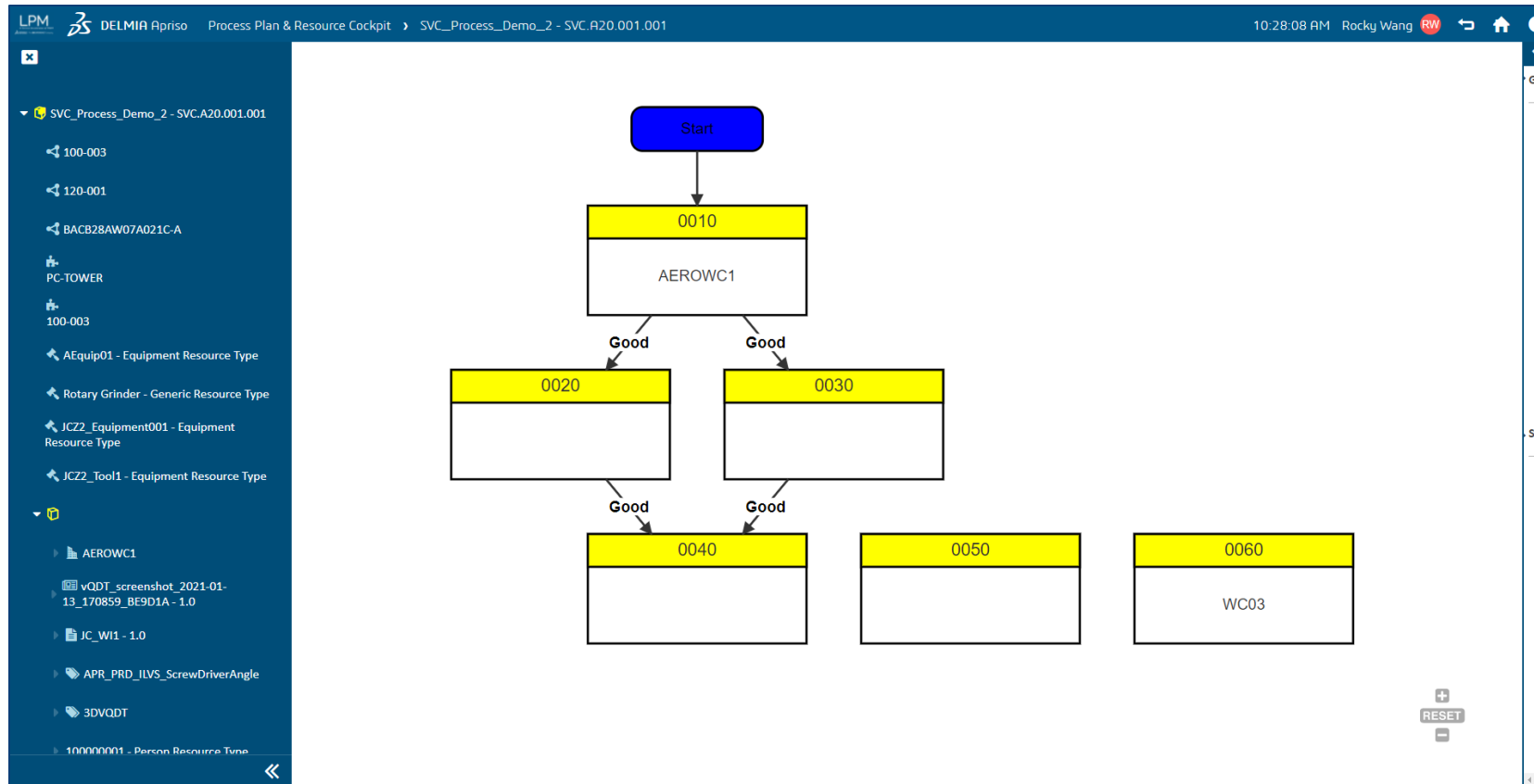
Download the PPR from DM and create order from downloaded PPR



- ▶ View Process definitions
- ▶ View components in the 3D Model
- ▶ Auto render to the Operation views in 3D Model
- ▶ View Process flow
- ▶ Show each item' detail information
- ▶ Digital Continuity with DM
- ▶ Create order by selected Process

PPR screen (Cont.)

View the Process definition from web screen instead of Process Builder



- ▶ View Process definitions
- ▶ View components in the 3D Model
- ▶ Auto render to the Operation views in 3D Model
- ▶ View Process flow
- ▶ Show each item' detail information
- ▶ Digital Continuity with DM
- ▶ Create order by selected Process

Enhanced Component view

LPM DELMIA Apriso Task List Operation Console 1:22:12 PM Rocky Wang RW

10704 Order
WING ASSY, UPPER, LIGHT WEIGHT Product

1 Operation
/ Serial/Lot

1 - LO Step Description
0 / 1 Qty

7/2/2021 7:00:00 AM Due Date

-99:99

0 / 1 (0%)

PartNo - Description	Instance	UOM	SerialNo/LotNo	Quantity	Total/ Kitted/ Remaining	Reference
444F5034-301 - 444F5034-301	444F5034-301.4	EA		Quantity	1/0/1	
444F5034-301 - 444F5034-301	444F5034-301.8	EA		Quantity	1/0/1	
NAS1399C4A4 - NAS1399C4A4	NAS1399C4A4.100	EA		Quantity	1/0/1	
NAS1399C4A4 - NAS1399C4A4	NAS1399C4A4.101	EA		Quantity	1/0/1	
NAS1399C4A4 - NAS1399C4A4	NAS1399C4A4.102	EA		Quantity	1/0/1	
NAS1399C4A4 - NAS1399C4A4	NAS1399C4A4.103	EA		Quantity	1/0/1	
NAS1399C4A4 - NAS1399C4A4	NAS1399C4A4.104	EA		Quantity	1/0/1	
NAS1399C4A4 - NAS1399C4A4	NAS1399C4A4.105	EA		Quantity	1/0/1	
NAS1399C4A4 - NAS1399C4A4	NAS1399C4A4.106	EA		Quantity	1/0/1	
NAS1399C4A4 - NAS1399C4A4	NAS1399C4A4.107	EA		Quantity	1/0/1	
NAS1399C4A4 - NAS1399C4A4	NAS1399C4A4.108	EA		Quantity	1/0/1	

Start Pause Report

Views

View.1

View.2

View.3

Selected component highlights on the 3D model

The component shows the installation location which defined in DM

Supports the Unload Components (service order)

The screenshot displays the DELMIA Apriso Operation Console interface. At the top, the header shows 'LPM', 'DELMIA Apriso', 'Task List', and 'Operation Console'. The main area is divided into several sections:

- Order Information:** Sample_JCZ22, Order, COMPONENT TRAY ASSY, UPPER WING, LH, Product.
- Operation Information:** 3 Operation, S7422 / Serial/Lot, 1 - ULOP, Step Description, 0 / 1 Qty.
- Due Date:** 8/3/2021 3:00:00 AM.
- Material Information:** S7419, MOM_PP_03TURBO, BOM, PL_02, VAL_FA_WS_1.

The central section is titled 'Unloading Components' and contains a table with the following data:

Product No	Serial No/Lot #	Quantity	Action
444F5034-301 -- (444F5034-301.4)		1.000	[Icon]
444F5034-301 -- (444F5034-301.8)		1.000	[Icon]
NAS1399C4A4 -- (NAS1399C4A4.229)		1.000	[Icon]
NAS1399C4A4 -- (NAS1399C4A4.230)		1.000	[Icon]
NAS1399C4A4 -- (NAS1399C4A4.231)		1.000	[Icon]
444F5027-301 -- (444F5027-301.4)		1.000	[Icon]

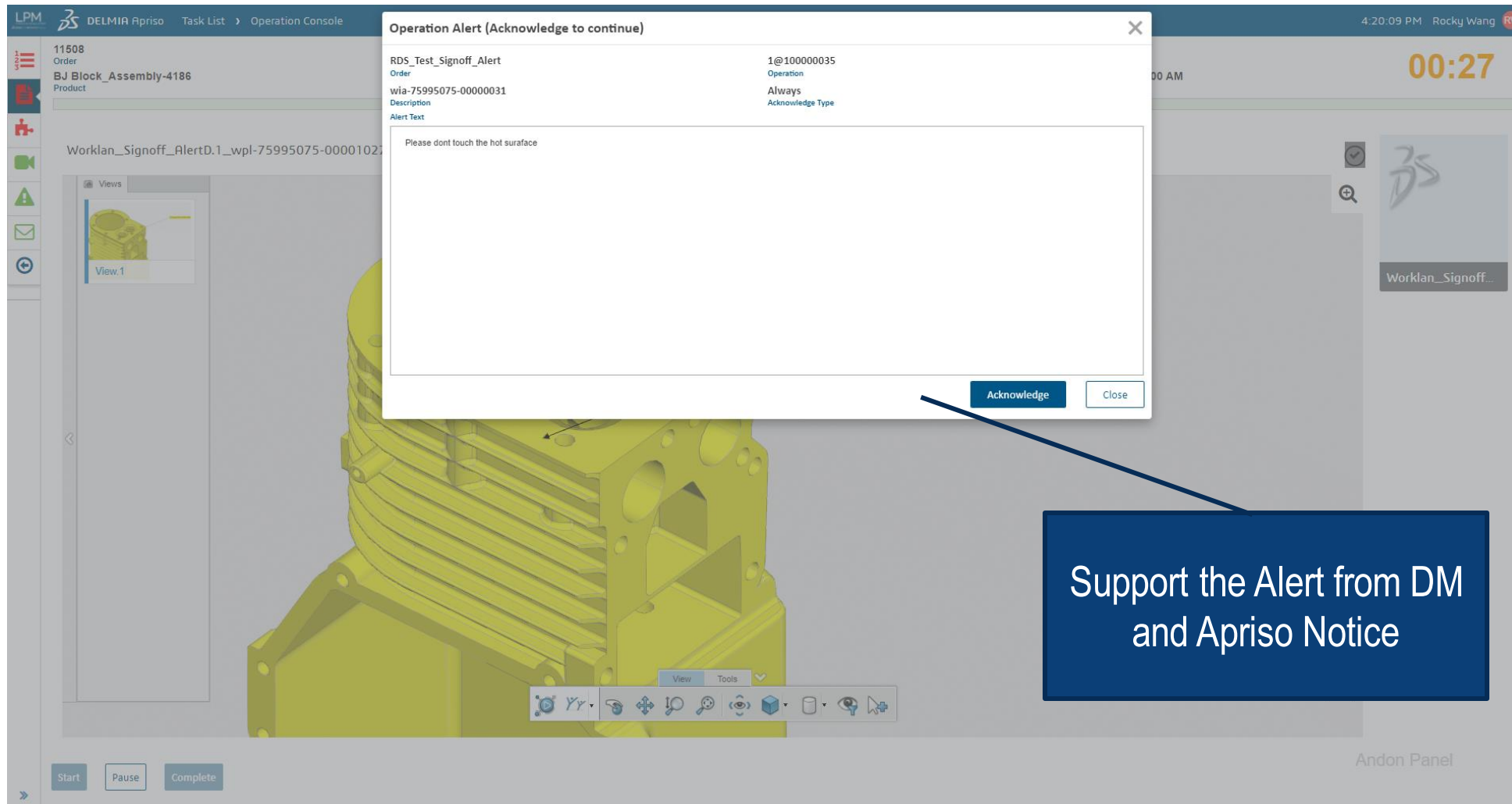
To the right of the table is a 3D model of a component, which is highlighted in blue. A blue arrow points from the first row of the table to the 3D model. Below the 3D model is a toolbar with various icons for viewing and interacting with the model. The text 'Andon Panel' is visible at the bottom right of the 3D model area.

At the bottom left of the interface, there are buttons for 'Start', 'Pause', and 'Complete'.

Selected component highlights on the 3D model

- Unload usage type is defined in DM or Apriso process or PPLS
- Auto to show selected unloading component in 3D Model (process from DM)

DM Alert and Apriso Notice



Signoff Step or Operation

10705
Order
120-001
Product

Task List > Operation Console

3/31/2021 2:00:00 PM Due Date

S77
120-001
201
Trim1_WS01

3:59:56 PM Rocky Wang

Signoff Required

Allow Partial SignOff: ☐ SignOff Non Mandatory: ☐

Instruction Text:
Signature for Wip Operation Signoff

Role/Actor: Production User
User: Rocky Password: *****
Comment:

Role/Actor: Portal Administrator
User: Rocky Password: *****
Comment:

Role/Actor: Production Supervisor
User: Rocky Password: *****
Comment:

SignOff Complete Cancel

History

Start Pause Complete

Andon Panel

Support the signoff definition from Dm and Apriso

Execute the Data Collect Plan from DM

LPM DELMIA Apriso Task List Operation Console 11:20:47 AM Rocky Wang RW

11509 Order 1 Operation 1 - General Operation00000426
PC-CASE Product S7462 / Serial/Lot 0 / 1 Qty 8/31/2021 7:00:00 AM Due Date
3 / 6 (50%)

-28:04

Data Collect Plan	Data Collect Name	UOM	Number Of Reads	Measurement	Last Value
widp-75995075-00000187	Data Collect Plan Row00000126		1/1	Char	
widp-75995075-00000187	Data Collect Plan Row00000127		1/1	Char	
widp-75995075-00000187	Data Collect Plan Row00000128	MMT	1/1	Calculate	

Data Collect Plan Row00000126

History

Start Pause Complete

Data Collect requirement from DM PPR

Work Instruction to collect data

New VQDT and 3DVQDT screen

The screenshot displays the LPM (Line Production Monitoring) software interface. The top header bar includes the LPM logo, 'DELMIA Apriso', 'Task List', and 'Operation Console'. The top right corner shows the time '10:58:32 AM' and the user 'Rocky Wang'. The main interface is divided into several sections:

- Order Information:** Sample_JCZ22, Order, COMPONENT TRAY ASSY, UPPER WING, LH, Product.
- Operation Information:** 3 Operation, S7422 / Serial/Lot.
- Step Information:** 1 - ULOP, Step Description, 0 / 1 Qty.
- Due Date:** 8/3/2021 3:00:00 AM.
- Timer:** S7419 MOM_PP_03TURBO... -99:99.

The central area features a table with columns: Defect #, Reason Code, Reference NCR #, Severity, No Of Defects, Status, and Comments. The table contains one row with the defect ID QD0001404, Reason Code Bent, Distord..., Severity Minor, No Of Defects 1, Status New, and a comment icon.

Below the table are buttons for 'Create NCR', 'Link NCR', and 'Unlink NCR'. To the right of the table is a 'REASON CODE' dropdown menu with options: Bent, Distorded, Broken, Carboned, Cosmetically Unacceptable, Cracked, Cross Threaded, Mechanical damage, Scratch, and Worn. Below this is a 'SEVERITY' dropdown menu with options: Minor, Major, and Critical.

The right side of the interface shows a 3D model of an aircraft wing with a red circle indicating a defect location. Below the model is a 'View' button and a 'Tools' button. The bottom right corner has a 'Start', 'Pause', and 'Complete' button.

- Interactive with operator
- Link reported Quality Defects with the NCR
- Work with NCR to close Quality Defects

Send NCR to the 3DExperience Platform

LPM DELMIA Apriso NCR List NCR Analysis / Repair 1:27:22 PM Rocky Wang

Analysis

Quality Defect

Order Number:	11201	Lot No:		Quantity:	1	Defect Details:	--
Product #:	120-001	Reject Code Type:	Defect Code	Date:	7/21/2021 10:28:31 PM	NCR Defect #:	QD0001402
Product Desc.:	RTR5 Test Product with Visualization	Reject Code:	Failure	Employee #:	Rocky		
Serial #:	S7412	Severity:	Normal	Work Center Found:	DispatchStation		

Assign to Department: Manufacturing Department *

Assign to Role: -- *

Assign to Employee: -- *

Disposition: Send to 3DEXP *

Analyst Comment: Change the routing and assign the components!

Attachments

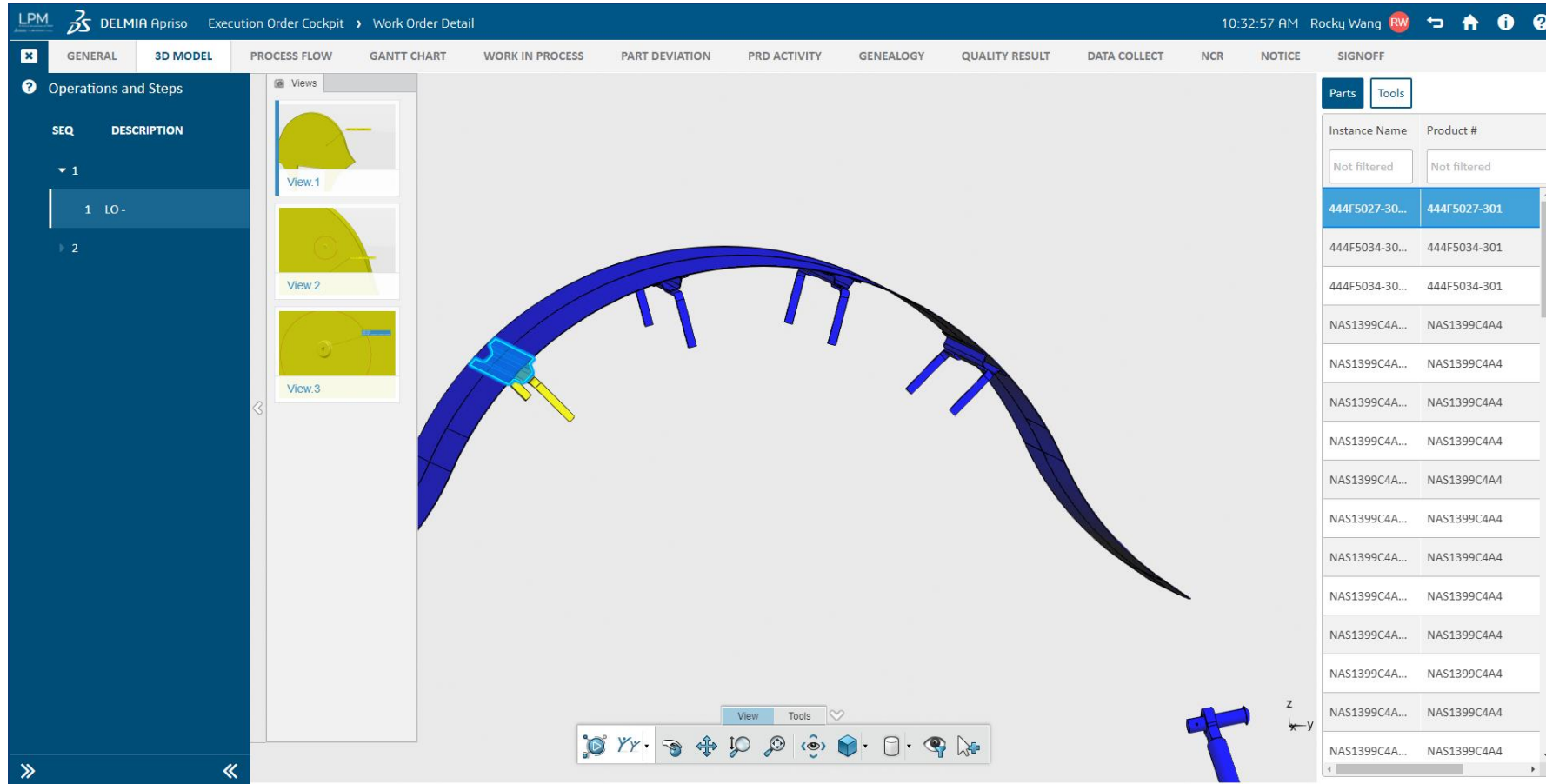
- 3DDriveLogo.jpg

Back Submit Analysis

Support the new
disposition – send the
NCR to 3D Experience
Platform

New Features of Work Order Cockpit

The 3D Model view for the order with 3D Model



- View the 3D Model of the linked process
- Auto render to the Operation views in 3D Model
- Select the components to show in 3D Model

New Features of Work Order Cockpit (Cont.)

Enhanced Work order cockpit to deliver the full information about the work order

The screenshot displays the 'Work Order Detail' page in the DELMIA Apriso Execution Order Cockpit. The interface includes a top navigation bar with tabs for INERAL, PROCESS FLOW, GANTT CHART, PART DEVIATION, PRD ACTIVITY (selected), GENEALOGY, QUALITY RESULT, DATA COLLECT, NCR, and NOTICE. A left sidebar shows a tree view of work orders, with '10003' expanded to show sub-items '0010', '0020', and '0030'. The main area features a table of activities with columns for Lot No, Activity, Activity Item01, and Activity Item02. The table lists various activities including 'operation', 'WO_ProgressStatus', 'ncr_rework_order_cr...', 'ncr_analysis', 'ncr_create', 'ncr_update', and 'Create3DVisualDefect'. A right panel displays key information: Wip Order No: 10003, Order Type: Production Order, Product No: KP_3D, Description: KP_3D, Operation Sequence: 0010, Step Sequence No: 0, Work Center: KP_3D_WS1, Serial No, Lot No, DefectNo: QD0001331, and Reason Code: DF-MEC-BNT. Below this information is a 3D model of a brake assembly with a red callout indicating a defect location. The bottom of the interface shows pagination controls and 'Selected records: 1'.

Lot No	Activity	Activity Item01	Activity Item02
Not filtered	Not filtered	Not filtered	Not filtered
	operation		21
	operation	1	0
	WO_ProgressStatus	WIP ORDER	1
	ncr_rework_order_cr...	8902	1
	ncr_analysis	QD0001332	
	ncr_create	QD0001332	
	ncr_update	QD0001229	
	ncr_update	QD0001229	
	Create3DVisualDefect	QD0001331	DF-MEC-BNT
	operation		21
	ncr_create	QD0001229	

► Full Activity history and detail information

► More information:

- ▷ Data Collection,
- ▷ NCR & Quality Defect
- ▷ Notice (DM alert)
- ▷ Signature

Genealogy Lookup and Work Order Activity

LPM DELMIA Apriso Genealogy Lookup 10:48:12 AM Rocky Wang RW

Product #: -- Serial #: S209 Lot No: Order Number: Filter Remove

SERIALNO-LOTNO-PRODUCTNO

- ▼ (S)S209-121-001
- (S)SN00001-120-001
- ▼ (L)LT0001-100-001
 - Parent
- ▼ (L)LT0002-100-002
 - Parent
- ▼ (L)LT0001-100-003
 - Parent
- 102-108
- 105-001
- 106-001
- ▼ (L)LT0001-110-001
 - Parent
- 110-002
- ▼ 110-003
 - Parent

GENERAL WORK IN PROCESS HISTORY PLANNED VS. ASBUILT

Product 121-001
 Product Description Body
 Product Revision 1.0
 Inventory Type Stock material
 Serial # S209

3D Model

Genealogy Detail

	Relation	Serial & Lot #	Product	Operation #	Step #	Quantity	UOM	Operation Status	Process
☐	Child	SN00001	120-001 - 1.0	1063	1	0010	2	1.000 EA	Started
☐	Child	LT0001	100-001 - 1.0	1063	1	0010	2	1.000 EA	Started
☐	Child	LT0002	100-002 - 1.0	1063	1	0010	2	1.000 EA	Started
☐	Child	LT0001	100-003 - 1.0	1063	1	0010	2	1.000 EA	Started
☐	Child		102-108 - 1.0	1063	1	0020	1	1.000 EA	Started
☐	Child		102-108 - 1.0	1063	1	0020	1	1.000 EA	Started
☐	Child		102-108 - 1.0	1063	1				Started
☐	Child		102-108 - 1.0	1063	1				Started
☐	Child		102-108 - 1.0	1063	1				Started

Selected records: 0

Filter to lookup

Show product image or 3D Model

Show unit genealogy (parent and children)

Genealogy Planned Vs. As Built

LPM

DELMIAPrismo

Genealogy Lookup

10:49:49 AM Rocky Wang

Product #: --

Serial #: S209

Lot No:

Order Number:

Filter

Remove

SERIALNO-LOTNO-PRODUCTNO

GENERAL

WORK IN PROCESS

HISTORY

PLANNED VS. ASBUILT

Genealogy As Planned vs. As Built

	Order Number	Operation #	Product #	Description	Type	Planned Quantity	As Built Quantity
	Not filtered	Not filtered	Not filtered	Not filtered	Not filtered	Not filtered	Not filtered
(S)S209-121-001 (S)SN00001-120-001 (L)LT0001-100-001 Parent (L)LT0002-100-002 Parent (L)LT0001-100-003 Parent 102-108 105-001 106-001 (L)LT0001-110-001 Parent 110-002 110-003 Parent							

1 / 1

Records: 14 Selected records: 0

LPM use cases

- ▶ Assembly Production
- ▶ Fabrication Production
- ▶ Repetitive Production
- ▶ Machine Driven Fast Production
- ▶ Nomad Production Use Case

LPM Update Agendas

► LPM Landscape

- ▷ LPM new Features
- ▷ PPR screen
- ▷ Execution
- ▷ NCR and Order Cockpit

► LPM Evolution / Roadmap

LPM Evolution

	2016	2017	2018	2019	2020	2021/2022
	PLS MBC	PPLS PLS MBC	NCR PRD Console PPLS PLS MBC	Andon Alerting Cockpit Wip Container Kitting Repetitive Fabrication Assembly	Automatic test Fast Production Nomad Planning Cockpit KPIs and Report Standard UI Client Mode Integration Ortems	3D Factory map Labeling Integration with WMS OOB REST Web Services PPR Viewer Digital continuity (DM)
		Lincoln Electric Bobcat Antoneum	Ingersoll Rand Pratt & Whitney Faraday Future	Carrier	PCC Sany (China) BW DAL (China)	LM
Version			A18	A19	A20	LPM.201 LPM.202